

Feedback, questions, and observations (living doc)

Canonical name: `20_feedback_questions_and_observations.md`

Purpose: Capture Charles’s reading-time questions and COMPASS answers. **Append new entries at the top** (newest first). Not part of the must-read print stack — use while working through `#01` – `#14` or inking Word.

PDF (you run locally — narrow margins): `./scripts/convert_single_md_to_pdf.sh 3-Master_Plan/20_feedback_questions_and_observations.md "" pdf_styles_narrow.css`
(COMPASS will not auto-export; re-run after `.md` edits when you want an updated print copy.)

Related: `Charles_reading_list.md` · `01_forward_plan_reading_guide.md`

How to add entries

```
### YYYY-MM-DD – Short title

**Context:** (which doc / section you were reading)

**Question / observation:** ...

**COMPASS answer:** ...
```

2026-06-11 — Full reading-stack shorthand scrub (all #01–#14)

Request: Audit every document on the print stack, not just `Charles_reading_list.md` — fix agent-only shorthands; explain B-lite origin.

COMPASS answer:

Term	Plain English
B-lite	COMPASS coinage (<code>#14</code> Q2, 2026-06-15): closure strong enough for Alex without full 3-domain identifiability. See <code>#01</code> “B-lite closure” section.
Path II	Manuscript-first lock: 3 empirical legs + 1 basketball generative POC + honest axis mismatch.
Rung 1–4	Paper-argument steps (phenomenon → mechanism → features → predictions → write). Rung 2.5 = model-guided measurements.

Files patched this pass: `#01` – `#09`, `#13`, `#14`, `#05`, `#10`, `#12` (+ prior `#20` § fixes). **Rule going forward:** first notebook mention = full path; capitalized agents need no gloss; `§` **default = manuscript Word** unless file says otherwise.

2026-06-11 — `§` symbol confusion + reading-list shorthand audit

Context: Cumulative “lost in the sauce” from agent-only shorthands (`§5`, bare `4D`, notebook numbers, etc.) while reading the must-read stack.

Question / observation: Does `**§N` always mean the manuscript? What else is hiding in the reading list that wasn’t designed to be forward-facing?

COMPASS answer: Usually yes — but not always. That ambiguity is a doc bug, not your fault.

Symbol	Default meaning	Common trap
<code>§N</code>	Word manuscript section (Dakota v03 / <code>#10</code>)	Agent note “§4 predictions” often meant staging <code>#12</code> §4 → manuscript §7, while manuscript §4 is tenure
<code>#NN</code>	Numbered reading file (<code>#07</code> , <code>#12</code> , ...)	<code>#12</code> §4 ≠ manuscript §4
<code>§X</code> in one file	Subsection inside that <code>.md</code> only	<code>#07</code> §F ≠ manuscript §F
530 / 537 / 538D / CELL 4D	Repo notebook labels — need full path on first mention	Bare “4D” is not four dimensions; it is <code>538D</code> CELL 4D

Audit done (this session):

- Added “**Symbol systems**” block to `01_forward_plan_reading_guide.md` — manuscript § map, staging `#12` § map, repo shorthand table.
- Updated `Charles_reading_list.md` decoder banner + clarified ink order (**manuscript §5** → **§1** → **§4**; predictions = **§7**).
- Fixed wrong “§4 Predictions” wording in this file (below) → **manuscript §7** / staging `#12` §4.

When stuck: open `#01` → **Symbol systems**; do not infer from context alone.

2026-07-01 — Notebook shorthand too bare (“4D” example)

Observation: In `#20` §5, the Wang-move hook row says “Near-threshold 4D (#1)” with nothing nearby saying **which notebook**, **which cell**, or **which export file**. With dozens of notebooks, that shorthand is one step too short.

COMPASS answer: Fair catch — agents know `538D` / `CELL 4D` from June context; you shouldn’t have to. **Rule:** any notebook/cell cite in `#20` or chat must include **path + cell label + export (if any)** on first mention. Expanded §5 below and `#01` glossary (`CELL 4D`, `538D`).

2026-07-01 — Shorthand lookup tax

Observation: Appreciate extensive shorthand for agent coordination, but ~1 in 7 times I still have to go looking.

COMPASS answer: Valid — shorthand is for **multi-agent lock-in**, not for making you decode every reply. **Going forward:** (1) first use of a term in chat gets a brief gloss or (#01) ; (2) file names when citing guidance, not bare “PD12”; (3) recurring misses get added to 01_forward_plan_reading_guide.md glossary (hook, PD12 guidance vs #14, POC, axis mismatch, scope discipline, sort-and-chop, overlap — added this session). **Flag any miss** in chat; treat it as a doc bug, not your fault.

2026-07-01 — Pool mean vs LOO; Wang predictions; Alex vs sorting; overlap; “hook”

Context: Reading #03 and related stack; random comments while working through must-read docs.

1. Pool mean vs LOO (Army + tenure) when AWS returns

Observation: Temporarily lost Army AWS access. When it returns, considering additional plots using **pool means** instead of LOOs. Same inverted-U on pool mean might save a justification step and simplify things — possibly for tenure too.

COMPASS answer: Worth doing as a supplementary check — not as a replacement for LOO in v1.

Why it could help: Generative POC reads out on **pool mean**; empirical v1 is locked on **LOO pool quality** (poolq_loo). That gap is the honest “axis mismatch” sentence in #05 . If Army (and maybe tenure) also show an inverted-U on a **whole-pool / team-mean** x-axis, you can say: *same qualitative shape on a second conditioning object*, which softens the “we only see it one way” worry.

Why it doesn’t fully remove the justification step:

Issue	LOO pool quality	Pool mean
Concept	“My peer environment excluding me ”	“My team’s average including me ”
Mechanism fit	Matches comparative evaluation / congestion story	Mechanically correlated with own ability
Alex / PD12	Primary empirical object	Secondary at best
Sorting discovery	About who is in the pool (overlap)	Different question — mean vs LOO is how you summarize the pool

Action when AWS returns: Run plots; label **supplementary**; compare side-by-side with LOO. If both bend the same way → nice Alex-meeting slide. If only LOO bends → **supports** comparative-evaluation framing. Either outcome is informative.

Tenure: poolq_loo = quality leg; pool_size_oa_loo = **congestion proxy**, not substitute for LOO quality. Pool-mean tenure plot → appendix / robustness, not main Setting 3 fact.

Do not delay manuscript inking for this. Tier 3 / post-AWS enrichment.

2. Doc #03 : “Non-obvious predictions distinguishing competing explanations”

Question: Is this a Wang reference? Explain, including Wang’s exact example.

COMPASS answer: Yes — **Wang-ladder language**. Contribution #3 in 03_Where_we_are_now.md : phenomenon → mechanism → **predictions that discriminate** → (deferred) network extensions.

“Wang move” in this project (#13 , #04):

A strong theory paper must **predict something you did not use to find the curve** — not only reproduce the inverted-U.

Inverted-U on poolq_loo = **Rung 1**. Wang-style paper also needs **Rung 3**: tests separating your story from cheaper alternatives (e.g. pure quadratic in team quality with **no** congestion channel).

This project’s two prediction slots:

#	Name	What it discriminates
#1	Near-threshold heterogeneity	Congestion hurts borderline elite , not superstars or non-contenders (Alex mechanism 3)
#2	Peak shift with Λ	Global slot scarcity moves the peak — finite distinction capacity vs purely local story

Wang’s example (Yin et al. 2019 — notes in 5-Manuscript/obsolete/superseded_briefings/wang_paper_model.md):

Not about inverted-U pools; the **logic** is what we imported.

- **Competing explanations beaten:** “Winners were always better” (raw early failure counts don’t separate); “Success is random / memoryless” (streaks aren’t memoryless).
- **Non-obvious prediction (k-memory model):** Actors reuse components from past attempts with memory **k**; critical **k*** below → stagnation, above → systematic improvement (phase transition). They test trajectory / learning patterns a failure-count model would not predict.

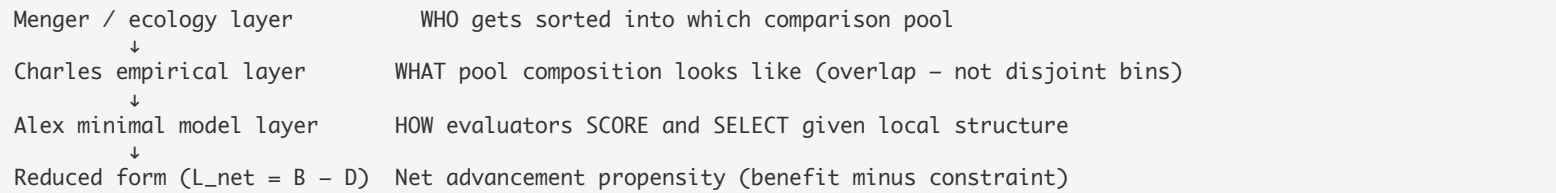
PD12 tie-in: Near-threshold harm $\approx$ NIH “**just at the cut line**” — substitutability among similar viable applicants.

Manuscript placement: Manuscript §7 (Predictions; ink from staging #12 §4) — inverted-U alone is not enough; #1 and #2 are **candidate** discriminators (honest language per #07).

3. Alex “score/selection” vs our “talent/sorting”

Observation: Gut feeling that Alex’s minimal model defines **selection / score / signal**, whereas our work defines **talent** and thus **sorting**. What is each model’s basic approach, how do they differ, how open to decomposition?

COMPASS answer: They sit at **different layers**; #05 is the nesting note.



	Alex (minimal model)	Charles emphasis (sorting + empirics)
Basic question	Given local peers, how does evaluation/selection work under congestion?	Given institutions, who ends up in which peer pool , and what does that do to advancement?
Core object	Score $S_i = A_i - \lambda C_{i,t} \rightarrow$ pick top K	Pool assignment + LOO pool quality $\rightarrow$ inverted-U
Congestion	Viable-peer density in selection rule	Also crowding_smooth , pool overlap, finite slots (Λ)
Generative POC	Soft assignment on fixed/simulated rosters; talent-only fails	538 pool assignment, overlap plots
Decomposition	Narrow: A vs D in score; six Alex mechanisms	Wider: B vs D in L_{net} ; sorting vs selection vs opportunity suppression
v1 role	Rung 2 POC + D-leg ontology	Rung 1 phenomenon + distinctive sorting/composition evidence

One sentence for Alex: *You supply the operational selection score and the quality-vs-congestion split; I supply cross-domain empirical regularity and evidence that realized pools are overlapping talent ecologies, not neat disjoint tiers.*

Not competing models if nesting chain is explicit. Alex is not explaining Duke's recruiting pipeline; we are not estimating six separate causal channels in one MBB table (PD12 scope discipline).

4. Sorting / pool overlap as top discovery (expanded)

Observation: Plots showing talent overlap and range of pools were highly informative — crushed naive model of sorting players 1–N and chopping into disjoint talent intervals as pool proxies.

COMPASS answer: This is one of your **strongest original empirical contributions**, separate from “we also see an inverted-U.” It answers a **prior** question that many papers skip: *before we interpret advancement vs local pool quality, what do comparison pools actually look like?*

The naive model (what we used to assume)

Early generative work — especially the frozen **537** “sort-and-chop” lab — treated pools like this:

1. Rank every player on ability ($1 \rightarrow N$).
2. Cut the continuum into **equal-count slices** (disjoint bins).
3. Call each slice a “pool” or team tier.

That assignment is **clean on paper** but **wrong on the data**. It produces interval “coverage” ≈ 1 everywhere on the performance axis: at any talent level, essentially **one** bin claims that player. No cross-pool sharing of talent levels.

What real pools look like (530 forensics — basketball)

530_sports_pipeline.ipynb CELLS 5–9 (documented in **sports/documents/Tier1_Presorting_Design_Note.md**) measure **realized** rosters, not simulated ones:

Finding	Numbers / pattern	Implication
Within-roster spread	Median roster SD $\approx 0.8\text{--}1.0$ z (PPM within-season)	Elite teams are not tight point masses of identical talent
Mean vs dispersion	Pearson r (team mean, roster SD) ≈ 0.26 — higher-mean teams slightly more dispersed, not less	Do not assume “elite $\Rightarrow$ homogeneous roster”
Interval overlap (CELL 8)	Actual team-season talent windows: peak coverage thousands of pools at $z \approx 0$	Many teams’ [min, max] intervals cross on the ability axis
Sort-and-chop benchmark	Coverage ≈ 1 everywhere (red dashed line in CELL 8)	The old 537 Assortative assignment matches red , not blue

Plain English: A player at a given talent level can plausibly appear on **many** kinds of teams. “Pool” is not a disjoint tier label — it is an **assortative but overlapping** comparison environment.

Same logic extends conceptually to Army units and tenure departments: local peer pools are **matched ecologies**, not non-overlapping bins on a global talent ladder.

Why this matters for the dissertation (beyond a methods footnote)

1. Legitimizes **LOO pool quality**. If pools were disjoint tiers, **poolq_loo** would collapse to “which bin am I in?” Overlap means LOO quality is a **continuous summary of a crowded peer field** — closer to Alex’s comparative-evaluation story.
2. Separates sorting from selection.
 - **Sorting (your layer):** *Who ends up in which overlapping pool?* (Menger **selective ecology / assortative matching**.)
 - **Selection (Alex layer):** *Given that local structure, how do evaluators score and pick?* ($S = A - \lambda C$.)The inverted-U on LOO quality is about **distinction inside** pools; overlap plots show **what those pools are made of**.

3. Motivates the generative fix (538 / `tier1_pool_assignment.py`). PD11 Thread A: draw team **target means** T_j , assign players with **soft** probabilities $\pi_{ij} \propto \exp(-(A_i - T_j)^2 / 2\tau^2)$ — overlap **by construction**. **538D** CELL 10 Plot A calibrates simulated overlap against 530 CELL 8; **537** **stays frozen** as the failed benchmark, not the forward path.
4. **Distinct from the inverted-U itself**. Reviewers could say “inverted-U is just curve-fitting.” Overlap forensics are harder to dismiss: they are a **structural fact about pool formation** that your old simulation explicitly got wrong and your new assignment explicitly fixes.

What to say vs not say (claim discipline)

Do say	Do not say
Realized comparison pools show substantial talent overlap across units/teams/depts	We have proven the soft-assignment generative process is the true DGP everywhere
Disjoint sort-and-chop is a useful null that fails on basketball forensics	Overlap plots replace the inverted-U or Alex POC
Overlap motivates assortative matching language (Menger) and honest LOO conditioning	Every domain has identical overlap statistics (Army forensics pending AWS)

Manuscript placement (concrete — manuscript § numbers)

Manuscript §	Content
§1 or §3 (empirical)	One overlap / coverage figure (<code>sports/530_sports_pipeline.ipynb</code> CELL 8 style, or cross-domain analog when Army returns) + paragraph: pools are overlapping, not disjoint tiers
§5 (generative / methods)	Brief contrast: talent-only fails (SCOUT C1/C2); sort-and-chop fails overlap ; soft assignment + congestion-in-score is the forward generative story
§6 or §5 discussion	Menger selective ecology — who gets matched into elite local environments
Appendix (optional)	<code>sports/537_tier1_benchmark.ipynb</code> vs 538D CELL 10 Plot A side-by-side for methods readers

Not a substitute for Alex's congestion-in-score POC — overlap answers *what we condition on*; Alex answers *how selection works given that conditioning*.

Decomposition chain (keep repeating this)

Sorting / ecology	→ overlapping pools (530 forensics; Menger Ch 10)
↓	
Local summary	→ LOO pool quality (poolq_loo) → inverted-U (Rung 1)
↓	
Selection / signal	→ Alex score, congestion, near-threshold predictions

One sentence for the paper: *Advancement under constrained distinction requires comparison pools that are assortative but overlapping; treating pools as disjoint talent bins misstates the peer environment and misaligns both our earliest simulations and any purely tier-based reading of the inverted-U.*

Repo pointers: **530** CELLS 5–9 · `Tier1_Presorting_Design_Note.md` · `538_Cell10_Generative_Manual.md` (Plot A) · `tier1_pool_assignment.py`

5. What does “hook” mean?

Question: Keep seeing “hook” — reminder? And what is the **spirit** of the word, not just the examples?

COMPASS answer:

Core definition (spirit of the word)

In this project, **hook** is workshop shorthand for an **attachment point** — a named place where an idea **hooks into** something else so it stays findable, honest, and actionable.

Not “marketing hook” (clickbait). Not “plot hook” (narrative suspense). Closer to:

A peg, handle, or anchor that ties an abstract claim, mechanism, or literature bridge to a **concrete landing zone** — manuscript section, repo file, companion chapter, or prediction name — **before** (or without) that landing zone being fully built or proven.

Three things a hook always does:

- Names where something belongs** so you and the agents don't re-litigate “where does Λ live?” every week.
- Admits incompleteness honestly** — a hook is allowed to be **prose-only**, **placeholder**, or **repo-target** until evidence arrives (**#07** guardrails).
- Keeps the stack connected** — theory ↔ measurement ↔ code ↔ chapter ↔ **manuscript §7** (predictions) stay linked without pretending the link is finished.

One-line definition: *A hook is a designated tether between an idea and where it will (or should) show up — not the proof itself.*

Why we say “hook” instead of “placeholder” or “anchor”

Word	Why “hook” fits better here
Placeholder	Sounds empty. A $\wedge$ hook still carries conceptual content (boards, draft picks, tenure slots) — you’re allowed to write manuscript §7 sentences about it (from staging #12 §4.2).
Anchor	Close, but static. “Hook” implies connection across layers (Menger ch → project story → Army prose).
TODO	Too task-list. Hooks are structural — part of the paper’s skeleton, not a stray chore.
Claim	Too strong. A hook is what you may point at before you may assert (see #07 : supported vs preliminary vs defer).

When docs say “ $\wedge$ hook prose,” they mean: *write the conceptual paragraph now; don’t claim the sweep figure exists.*

What a hook is not

Not a hook	Why
A finished result	D10 near-threshold export is evidence; “near-threshold hook” was the named slot before the PNG existed.
A vague vibe	“Congestion matters” is a theme. “ Repo hook : tier1_pool_assignment.py ” is a hook — you know where to implement.
Permission to overclaim	Hooks pair with claim discipline. “Conceptual hook” $\neq$ “demonstrated in all three domains.”
The inverted-U itself	Rung 1 phenomenon is found ; hooks are mostly Rung 3+ or infrastructure (where predictions / code / lit attach).

Four usages in project docs (examples)

Usage	What it hooks from → to	Example
Prediction hook	Mechanism → manuscript §7 prose (figure optional)	“ $\wedge$ hook ” = global slot capacity shifts peak (prediction #2); Army-led prose in §7 — no finished $\wedge$ -sweep figure in repo yet
Wang-move hook	Model logic → test beyond discovery curve	Prediction #1 near-threshold heterogeneity — see worked example below (not bare “4D”)
Repo / PD12 hook	Priority / mechanism → codebase location	“Repo hook: sports/539_alex_model.ipynb ” in 20260520_Transcript_12_guidance.md
Literature hook	Companion chapter → your empirical story	Menger calibration table “project hook” column (Ch 10 → selective ecology / overlap)

Worked example — Prediction #1 (Wang-move hook, was “near-threshold 4D”)

What	Where
Prediction	Near-threshold heterogeneity — congestion should bite borderline elite , not superstars or non-contenders (discriminates congestion from “elite pools are just hard”)
Notebook	sports/538D_development.ipynb
Cell	CELL 4D (fourth sub-block under CELL 4 — letter D , not “four dimensions”)
Script	sports/tier1_heterogeneity_ventiles.py
Export (D10)	datasets/mbb/exports_inverted_u_v0/scout_manuscript_v1/heterogeneity_ventiles_top_tail.png (list in #08)
Manuscript	Manuscript §7 (Predictions; ink from staging #12 §4) — candidate readout per #07 (exploratory, not fully validated cross-domain)

Do not write “4D” alone in prose meant for you — always 538D **CELL 4D** or the **export filename** the first time.

How to hear it in conversation

- “What’s the $\wedge$ **hook**?” → *Where in the manuscript and which domain leads the argument?* (Answer: **manuscript §7**, Army prose, boards/slots — not “we simulated $\wedge$ in 538D CELL 10.”)

- “Is there a **repo hook** for near-threshold?” → *Which notebook/file owns that readout?* (`sports/538D_development.ipynb` CELL 4D → `heterogeneity_ventiles_top_tail.png` in D10 bundle — `#08` .)
- “Menger **hook** for overlap?” → *Which companion chapter justifies assortative-but-overlapping pools?* (Selective ecology / matching — `#20260617` calibration doc.)

Mnemonic: Hook = where it *hangs*. Proof = what *holds* it up.

End entry 2026-07-01.